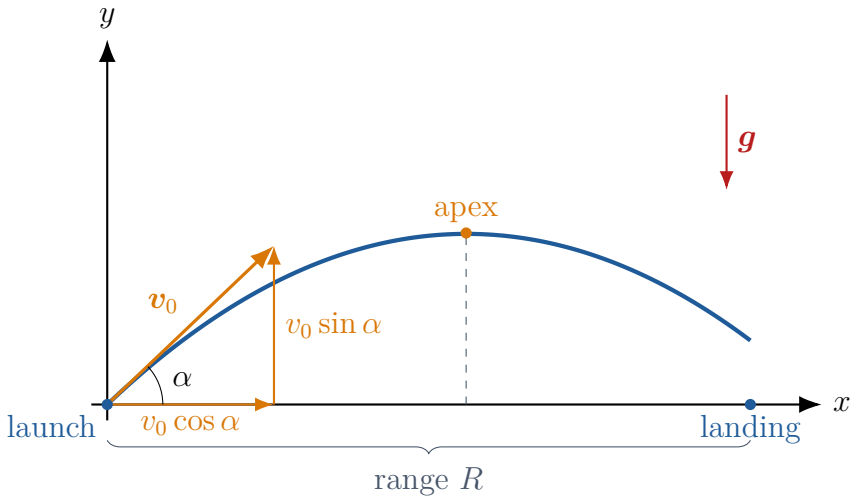
Projectile Motion

lecture notes with full derivations



Ideal model: no air resistance, constant gravitational acceleration, motion in one vertical plane.

A study-oriented handout: physical assumptions, equations, derivations, examples, typical errors, and practice problems.

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1 Purpose and physical model

Projectile motion is the motion of a body that is launched with an initial velocity not necessarily parallel to the ground and then moves only under the influence of gravity. It is one of the most useful introductory models in mechanics because it shows how a simple force law leads to concrete predictions: the shape of the path, the time of flight, the maximum height, and the horizontal range.

Assumptions of the ideal projectile model

Throughout these notes, unless stated otherwise, we assume:

1. the body can be treated as a point particle;
2. air resistance is neglected;
3. the gravitational acceleration is constant and vertical;
4. the motion takes place in one vertical plane, described by axes x and y;
5. the x-axis is horizontal and the y-axis is vertical, positive upward.

Common notation:

Symbol	Meaning
v_0	magnitude of the initial velocity
α	launch angle measured from the horizontal
g	magnitude of gravitational acceleration, usually $g \approx 9.81\,\mathrm{m/s^2}$
y_0	initial height above the chosen reference level
R	horizontal range, i.e. horizontal distance from launch to landing
$H_{\max}$	maximum height above the chosen reference level

1.1 Why the motion separates into horizontal and vertical parts

The key idea is that horizontal and vertical coordinates can be analyzed separately because gravity has only a vertical component. This does not mean that two different objects are moving. There is one motion, but it is described by two coordinate functions.

The position vector is

$$\boldsymbol{r}(t) = x(t)\,\boldsymbol{e}_x + y(t)\,\boldsymbol{e}_y.$$

Velocity and acceleration are derivatives with respect to time:

$$\boldsymbol{v}(t) = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \dot{x}(t)\,\boldsymbol{e}_x + \dot{y}(t)\,\boldsymbol{e}_y,$$

$$\boldsymbol{a}(t) = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \ddot{x}(t)\,\boldsymbol{e}_x + \ddot{y}(t)\,\boldsymbol{e}_y.$$

For ideal projectile motion the only force is the weight:

$$\boldsymbol{F} = m\boldsymbol{g} = -mg\,\boldsymbol{e}_y.$$

Newton's second law $\boldsymbol{F} = m\boldsymbol{a}$ gives

$$m(\ddot{x}\,\boldsymbol{e}_x + \ddot{y}\,\boldsymbol{e}_y) = -mg\,\boldsymbol{e}_y.$$

Comparing components gives the system

$$\boxed{\ddot{x} = 0}, \qquad \boxed{\ddot{y} = -g}.$$

Therefore the horizontal coordinate has zero acceleration, while the vertical coordinate has constant acceleration downward.

Most important conceptual point

The horizontal and vertical motions share the same time variable t. We may solve them separately, but they occur simultaneously. The projectile does not first move horizontally and then vertically; both coordinates change at the same time.

2 Initial velocity and component form

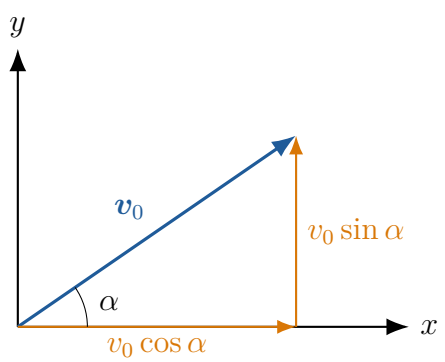
Suppose the projectile is launched from the point (x_0, y_0) with initial speed v_0 at angle α above the horizontal.

The initial velocity vector is decomposed as

$$\boldsymbol{v}(0) = v_0 \cos\alpha\,\boldsymbol{e}_x + v_0 \sin\alpha\,\boldsymbol{e}_y.$$

Thus

$$v_{0x} = v_0 \cos\alpha, \qquad v_{0y} = v_0 \sin\alpha.$$



Component interpretation

$$v_{0x} = v_0 \cos\alpha \quad \text{controls horizontal motion,}$$

$$v_{0y} = v_0 \sin\alpha \quad \text{controls vertical motion.}$$

For the ideal model, v_x remains constant, while v_y changes linearly because of gravity.

3 Derivation of the equations of motion

The component equations are

$$\ddot{x} = 0, \qquad \ddot{y} = -g.$$

We now integrate them with the initial conditions

$$x(0) = x_0, \qquad y(0) = y_0,$$
$$\dot{x}(0) = v_0 \cos\alpha, \qquad \dot{y}(0) = v_0 \sin\alpha.$$

Horizontal equation

Start from

$$\ddot{x} = 0.$$

Integrate once with respect to time:

$$\dot{x}(t) = C_1.$$

Use the initial horizontal velocity:

$$\dot{x}(0) = v_{0x} = v_0 \cos\alpha.$$

Therefore

$$\dot{x}(t) = v_0 \cos\alpha.$$

Integrate again:

$$x(t) = v_0 \cos\alpha\, t + C_2.$$

Use $x(0) = x_0$:

$$x_0 = C_2.$$

Hence

$$\boxed{x(t) = x_0 + v_0 \cos\alpha\, t.}$$

Vertical equation

Start from

$$\ddot{y} = -g.$$

Integrate once:

$$\dot{y}(t) = -gt + C_3.$$

Use the initial vertical velocity:

$$\dot{y}(0) = v_{0y} = v_0 \sin\alpha,$$

so

$$C_3 = v_0 \sin\alpha.$$

Thus

$$\dot{y}(t) = v_0 \sin\alpha - gt.$$

Integrate again:

$$y(t) = v_0 \sin\alpha\, t - \frac{1}{2}gt^2 + C_4.$$

Use $y(0) = y_0$:

$$C_4 = y_0.$$

Therefore

$$\boxed{y(t) = y_0 + v_0 \sin\alpha\, t - \frac{1}{2}gt^2.}$$

Together, the parametric equations of ideal projectile motion are

Parametric equations

$$\boxed{x(t) = x_0 + v_0 \cos\alpha\, t,}$$

$$\boxed{y(t) = y_0 + v_0 \sin\alpha\, t - \frac{1}{2}gt^2.}$$

The velocity components are

$$\boxed{v_x(t) = v_0 \cos\alpha,} \qquad \boxed{v_y(t) = v_0 \sin\alpha - gt.}$$

3.1 Meaning of each term

In the vertical equation

$$y(t) = y_0 + v_0 \sin\alpha\, t - \frac{1}{2}gt^2,$$

the term y_0 is the initial height, $v_0 \sin\alpha\, t$ is the upward displacement that would occur if there were no gravity, and $-\frac{1}{2}gt^2$ is the downward displacement caused by gravitational acceleration.

In the horizontal equation

$$x(t) = x_0 + v_0 \cos\alpha\, t,$$

there is no quadratic term because, in the ideal model, no horizontal force acts on the projectile.

4 The shape of the trajectory

The parametric equations describe the motion using time as a parameter. To find the geometric shape of the path, eliminate time.

For simplicity take $x_0 = 0$. Then

$$x(t) = v_0 \cos\alpha\, t.$$

If $\cos\alpha \neq 0$, then

$$t = \frac{x}{v_0 \cos\alpha}.$$

Substitute this into the vertical equation:

$$y = y_0 + v_0 \sin\alpha \left(\frac{x}{v_0 \cos\alpha}\right) - \frac{1}{2} g \left(\frac{x}{v_0 \cos\alpha}\right)^2.$$

Simplify the first nonconstant term:

$$v_0 \sin\alpha \left(\frac{x}{v_0 \cos\alpha}\right) = x \tan\alpha.$$

Therefore

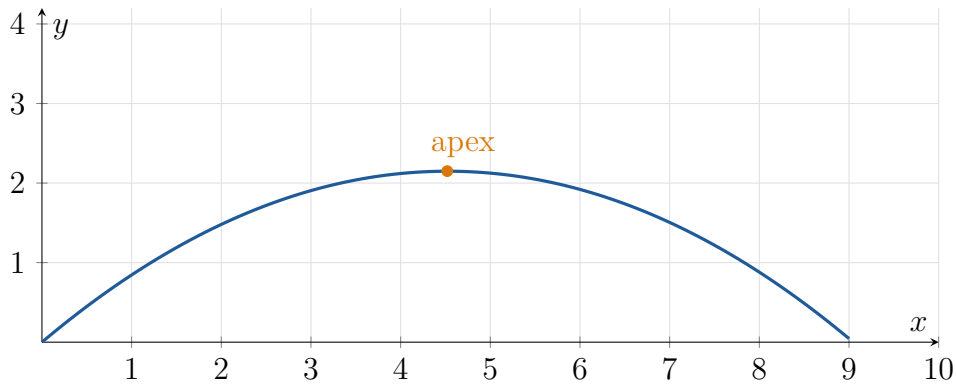
$$\boxed{y(x) = y_0 + x\tan\alpha - \frac{g}{2v_0^2 \cos^2\alpha} x^2.}$$

Why the path is a parabola

The trajectory has the form

$$y(x) = A + Bx - Cx^2,$$

where $C > 0$. Therefore it is a downward-opening parabola. The parabolic shape follows from constant vertical acceleration and zero horizontal acceleration.



5 Velocity, speed, and acceleration during flight

The velocity vector is

$$\boldsymbol{v}(t) = v_0 \cos\alpha\, \boldsymbol{e}_x + (v_0 \sin\alpha - gt)\, \boldsymbol{e}_y.$$

The acceleration vector is constant:

$$\boldsymbol{a}(t) = -g\, \boldsymbol{e}_y.$$

The speed is the magnitude of the velocity:

$$v(t) = \|\boldsymbol{v}(t)\| = \sqrt{(v_0 \cos\alpha)^2 + (v_0 \sin\alpha - gt)^2}.$$

5.1 The velocity at the highest point

At the highest point the vertical component of the velocity is zero. This is not the same as saying that the whole velocity is zero. Unless the motion is purely vertical, the horizontal component remains nonzero.

At the top:

$$v_y = 0, \qquad v_x = v_0 \cos\alpha.$$

Therefore the speed at the top is

$$\boxed{v_{\text{top}} = v_0 \cos\alpha.}$$

A common misconception

At the maximum height, the projectile is not at rest. Only the vertical component of its velocity is zero. The horizontal component is still present, so the projectile continues to move horizontally.

6 Maximum height

We now derive the maximum height. For clarity, we first assume the projectile starts at ground level, so $y_0 = 0$. Then we state the general result for arbitrary initial height.

6.1 Time to reach the highest point

The vertical velocity is

$$v_y(t) = v_0 \sin\alpha - gt.$$

At the maximum height,

$$v_y(t_{\text{max}}) = 0.$$

Thus

$$v_0 \sin\alpha - gt_{\text{max}} = 0,$$

so

$$\boxed{t_{\text{max}} = \frac{v_0 \sin\alpha}{g}.}$$

6.2 Height at the highest point

For $y_0 = 0$,

$$y(t) = v_0 \sin\alpha\, t - \frac{1}{2}gt^2.$$

Substitute $t = t_{\text{max}} = \frac{v_0 \sin\alpha}{g}$:

$$H_{\text{max}} = v_0 \sin\alpha \left(\frac{v_0 \sin\alpha}{g}\right) - \frac{1}{2}g\left(\frac{v_0 \sin\alpha}{g}\right)^2.$$

Compute each term:

$$v_0 \sin\alpha \left(\frac{v_0 \sin\alpha}{g}\right) = \frac{v_0^2 \sin^2\alpha}{g},$$

$$\frac{1}{2} g \left(\frac{v_0 \sin\alpha}{g}\right)^2 = \frac{v_0^2 \sin^2\alpha}{2g}.$$

Therefore

$$\boxed{H_{\max} = \frac{v_0^2 \sin^2\alpha}{2g}.}$$

If the projectile starts at height y_0, then the maximum height above the reference level is

$$\boxed{H_{\max} = y_0 + \frac{v_0^2 \sin^2\alpha}{2g}.}$$

Energy derivation of maximum height

The vertical part of the initial kinetic energy is converted into gravitational potential energy during the upward part of the motion. If the launch height is taken as zero, then

$$\frac{1}{2} m (v_0 \sin\alpha)^2 = mgH_{\max}.$$

Cancel m:

$$\frac{1}{2} v_0^2 \sin^2\alpha = gH_{\max}.$$

Hence

$$H_{\max} = \frac{v_0^2 \sin^2\alpha}{2g}.$$

This agrees with the kinematic derivation.

7 Time of flight and range: launch and landing at the same height

Assume now that the projectile is launched and lands at the same vertical level. Take $y_0 = 0$ and landing at $y = 0$.

The vertical equation is

$$y(t) = v_0 \sin\alpha\, t - \frac{1}{2} g t^2.$$

At landing,

$$y(t) = 0.$$

Therefore

$$v_0 \sin\alpha\, t - \frac{1}{2} g t^2 = 0.$$

Factor t:

$$t \left(v_0 \sin\alpha - \frac{1}{2} g t\right) = 0.$$

This gives two roots:

$$t = 0 \quad \text{and} \quad t = \frac{2v_0 \sin\alpha}{g}.$$

The first root is the launch instant. The nonzero root is the flight time:

$$\boxed{t_{\text{flight}} = \frac{2v_0 \sin\alpha}{g}.}$$

The horizontal range is horizontal speed times flight time:

$$R = (v_0 \cos\alpha) t_{\text{flight}}.$$

Substitute the expression for t_{flight}:

$$R = v_0 \cos\alpha \cdot \frac{2v_0 \sin\alpha}{g} = \frac{2v_0^2 \sin\alpha \cos\alpha}{g}.$$

Use the identity

$$\sin(2\alpha) = 2\sin\alpha\cos\alpha.$$

Thus

$$\boxed{R = \frac{v_0^2 \sin(2\alpha)}{g}.}$$

Same launch and landing height

If launch and landing occur at the same height, then

$$\boxed{t_{\text{flight}} = \frac{2v_0 \sin\alpha}{g}}, \qquad \boxed{R = \frac{v_0^2 \sin(2\alpha)}{g}}.$$

These formulas are not valid without modification when the landing height differs from the launch height.

8 Why 45 degrees gives maximum range

For fixed v_0 and g, the range for equal launch and landing height is

$$R(\alpha) = \frac{v_0^2}{g} \sin(2\alpha).$$

The factor v_0^2/g is constant, so maximizing R is the same as maximizing $\sin(2\alpha)$.

For $0 \le \alpha \le 90°$, we have

$$0 \le 2\alpha \le 180°.$$

The sine function reaches its maximum value 1 at

$$2\alpha = 90°.$$

Therefore

$$\boxed{\alpha = 45^\circ}$$

gives the maximum range, and the maximum range is

$$\boxed{R_{\max} = \frac{v_0^2}{g}.}$$

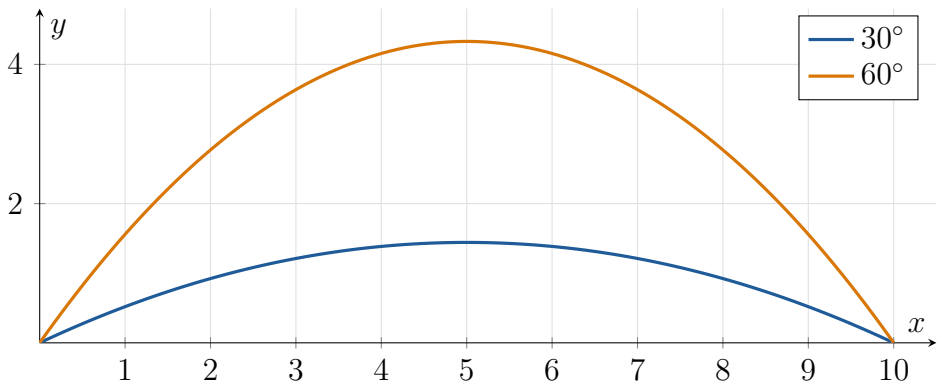
8.1 Complementary launch angles

Because

$$\sin(2\alpha) = \sin(180^\circ - 2\alpha),$$

angles α and $90^\circ - \alpha$ give the same range when launch and landing heights are equal.

For example, 30° and 60° give the same range. The 60° trajectory has a larger maximum height and a longer time of flight; the 30° trajectory is flatter and faster horizontally.



9 Projectile launched from a height

In many problems the projectile is launched from a height y_0 above the ground. Suppose the ground is $y = 0$, the launch point is $(0, y_0)$, and the projectile lands when $y = 0$.

The vertical equation is

$$y(t) = y_0 + v_0 \sin\alpha\, t - \frac{1}{2}gt^2.$$

At landing,

$$y_0 + v_0 \sin\alpha\, t - \frac{1}{2}gt^2 = 0.$$

Rearrange as a standard quadratic equation:

$$\frac{1}{2}gt^2 - v_0 \sin\alpha\, t - y_0 = 0.$$

Using the quadratic formula with

$$a = \frac{1}{2}g, \qquad b = -v_0 \sin\alpha, \qquad c = -y_0,$$

we obtain

$$t = \frac{v_0 \sin\alpha \pm \sqrt{v_0^2 \sin^2\alpha + 2gy_0}}{g}.$$

The physically relevant landing time is the positive root:

$$\boxed{t_{\text{flight}} = \frac{v_0 \sin\alpha + \sqrt{v_0^2 \sin^2\alpha + 2gy_0}}{g}.}$$

Then the range is

$$\boxed{R = v_0 \cos\alpha \, t_{\text{flight}}.}$$

Do not use the equal-height formula blindly

The formula

$$R = \frac{v_0^2 \sin(2\alpha)}{g}$$

assumes that launch and landing occur at the same height. If the projectile is thrown from a cliff or a table, use the vertical equation to find the actual flight time first.

10 Projectile aimed at a target

Suppose a projectile is launched from $(0, y_0)$ with speed v_0 and angle α. We want to know whether it can pass through a target point (X, Y).

The trajectory equation is

$$y(x) = y_0 + x\tan\alpha - \frac{g}{2v_0^2\cos^2\alpha}x^2.$$

Put $x = X$ and $y = Y$:

$$Y = y_0 + X\tan\alpha - \frac{gX^2}{2v_0^2\cos^2\alpha}.$$

This equation can be solved for α. It is useful to write it in terms of

$$u = \tan\alpha.$$

Use the identity

$$\frac{1}{\cos^2\alpha} = 1 + \tan^2\alpha = 1 + u^2.$$

Then

$$Y = y_0 + Xu - \frac{gX^2}{2v_0^2}(1 + u^2).$$

Move all terms to one side:

$$\frac{gX^2}{2v_0^2}u^2 - Xu + \left(Y - y_0 + \frac{gX^2}{2v_0^2}\right) = 0.$$

This is a quadratic equation in $u = \tan\alpha$.

Interpretation of the two angles

If this quadratic equation has two positive real solutions, there are two possible launch angles: a low, flat trajectory and a high, steep trajectory. If it has one solution, the target is reached at the limiting angle. If it has no real solution, the speed v_0 is too small for the chosen target.

10.1 Minimum launch speed for reaching a point

A compact way to identify whether a point can be reached is to use the envelope of trajectories. For a fixed launch speed v_0 and launch point at $(0, 0)$, the family of trajectories is

$$y = x \tan \alpha - \frac{gx^2}{2v_0^2 \cos^2 \alpha}.$$

Set $u = \tan \alpha$ and use $1/\cos^2 \alpha = 1 + u^2$:

$$y = xu - \frac{gx^2}{2v_0^2}(1 + u^2).$$

For fixed x, this is a quadratic function of u:

$$y(u) = -\frac{gx^2}{2v_0^2}u^2 + xu - \frac{gx^2}{2v_0^2}.$$

The greatest possible height at that horizontal position is the maximum of this quadratic. The maximum occurs when

$$\frac{\mathrm{d}y}{\mathrm{d}u} = -\frac{gx^2}{v_0^2}u + x = 0.$$

Thus

$$u = \frac{v_0^2}{gx}.$$

Substitute into $y(u)$:

$$y_{\text{env}}(x) = \frac{v_0^2}{2g} - \frac{gx^2}{2v_0^2}.$$

Therefore a point (x, y) can be reached only if it lies at or below

$$\boxed{y_{\text{env}}(x) = \frac{v_0^2}{2g} - \frac{gx^2}{2v_0^2}.}$$

11 Impact velocity and impact angle

The impact velocity is found by evaluating the velocity components at the landing time.

The components are

$$v_x(t) = v_0 \cos \alpha, \qquad v_y(t) = v_0 \sin \alpha - gt.$$

If the landing time is t_{flight}, then

$$v_{x,\text{impact}} = v_0 \cos\alpha,$$

$$v_{y,\text{impact}} = v_0 \sin\alpha - g t_{\text{flight}}.$$

The impact speed is

$$\boxed{v_{\text{impact}} = \sqrt{v_{x,\text{impact}}^2 + v_{y,\text{impact}}^2}.}$$

The impact angle β measured below the horizontal satisfies

$$\boxed{\tan\beta = \frac{|v_{y,\text{impact}}|}{v_{x,\text{impact}}}.}$$

For equal launch and landing heights, the vertical impact velocity is

$$v_{y,\text{impact}} = -v_0 \sin\alpha.$$

Therefore the impact speed equals the launch speed:

$$v_{\text{impact}} = \sqrt{(v_0\cos\alpha)^2 + (v_0\sin\alpha)^2} = v_0.$$

This is also expected from conservation of mechanical energy.

12 Energy viewpoint

The kinematic equations describe position and velocity as functions of time. Energy provides a complementary way to understand speed at a given height.

In the ideal model, mechanical energy is conserved:

$$\frac{1}{2}mv^2 + mgy = \frac{1}{2}mv_0^2 + mgy_0.$$

Cancel m:

$$\frac{1}{2}v^2 + gy = \frac{1}{2}v_0^2 + gy_0.$$

Solve for v^2:

$$v^2 = v_0^2 + 2g(y_0 - y).$$

Thus

$$\boxed{v(y) = \sqrt{v_0^2 + 2g(y_0 - y)}.}$$

What energy can and cannot tell you

Energy gives the speed as a function of height, but not the direction of the velocity. To get the direction, you still need the velocity components or the trajectory geometry.

13 Dimensional checks

Checking units is one of the fastest ways to detect algebraic mistakes.

For time of flight at equal height:

$$t_{\text{flight}} = \frac{2v_0 \sin\alpha}{g}.$$

Units:

$$\frac{\text{m/s}}{\text{m/s}^2} = \text{s}.$$

Correct.

For maximum height:

$$H_{\max} = \frac{v_0^2 \sin^2\alpha}{2g}.$$

Units:

$$\frac{\text{m}^2/\text{s}^2}{\text{m/s}^2} = \text{m}.$$

Correct.

For range:

$$R = \frac{v_0^2 \sin(2\alpha)}{g}.$$

Units:

$$\frac{\text{m}^2/\text{s}^2}{\text{m/s}^2} = \text{m}.$$

Correct.

14 Worked examples

Example 1: basic projectile from ground level

A ball is launched from ground level with speed $v_0 = 20\,\text{m/s}$ at angle $\alpha = 30°$. Find the time of flight, maximum height, and range. Use $g = 9.81\,\text{m/s}^2$.

Solution. For equal launch and landing height,

$$t_{\text{flight}} = \frac{2v_0 \sin\alpha}{g} = \frac{2 \cdot 20 \cdot \sin 30°}{9.81} = \frac{20}{9.81} \approx 2.04\,\text{s}.$$

The maximum height is

$$H_{\max} = \frac{v_0^2 \sin^2\alpha}{2g} = \frac{20^2 \cdot (1/2)^2}{2 \cdot 9.81} = \frac{100}{19.62} \approx 5.10\,\text{m}.$$

The range is

$$R = \frac{v_0^2 \sin(2\alpha)}{g} = \frac{400 \sin 60°}{9.81} \approx \frac{400 \cdot 0.866}{9.81} \approx 35.3\,\text{m}.$$

Example 2: projectile launched from a height

A stone is thrown from a height $y_0 = 15\,\mathrm{m}$ with speed $v_0 = 12\,\mathrm{m/s}$ at angle $\alpha = 25°$ above the horizontal. Find the flight time and range.

Solution. The vertical equation is

$$0 = 15 + 12\sin 25°\, t - \frac{1}{2}(9.81)t^2.$$

The positive root is

$$t_{\text{flight}} = \frac{12\sin 25° + \sqrt{12^2\sin^2 25° + 2\cdot 9.81\cdot 15}}{9.81}.$$

Using $\sin 25° \approx 0.423$,

$$t_{\text{flight}} \approx \frac{5.07 + \sqrt{25.7 + 294.3}}{9.81} = \frac{5.07 + 17.89}{9.81} \approx 2.34\,\mathrm{s}.$$

The range is

$$R = 12\cos 25° \cdot 2.34.$$

Using $\cos 25° \approx 0.906$,

$$R \approx 12 \cdot 0.906 \cdot 2.34 \approx 25.4\,\mathrm{m}.$$

Example 3: finding the velocity after a given time

A projectile is launched with speed $18\,\mathrm{m/s}$ at $40°$. Find the velocity vector after $1.0\,\mathrm{s}$.

Solution. The horizontal component is constant:

$$v_x = 18\cos 40° \approx 13.8\,\mathrm{m/s}.$$

The vertical component is

$$v_y = 18\sin 40° - 9.81\cdot 1.0 \approx 11.6 - 9.81 = 1.79\,\mathrm{m/s}.$$

So

$$\boldsymbol{v}(1.0\,\mathrm{s}) \approx (13.8\,\boldsymbol{e}_x + 1.79\,\boldsymbol{e}_y)\,\mathrm{m/s}.$$

The projectile is still moving upward, because $v_y > 0$.

Example 4: angle needed to hit a point

A projectile is launched from the origin with speed $v_0 = 25\,\mathrm{m/s}$. It must pass through a point $X = 40\,\mathrm{m}$, $Y = 5\,\mathrm{m}$. Find possible launch angles.

Solution. Use

$$Y = Xu - \frac{gX^2}{2v_0^2}(1 + u^2), \qquad u = \tan\alpha.$$

Substitute the numbers:

$$5 = 40u - \frac{9.81\cdot 40^2}{2\cdot 25^2}(1 + u^2).$$

Compute the coefficient:

$$\frac{9.81 \cdot 1600}{1250} \approx 12.56.$$

Thus

$$5 = 40u - 12.56(1 + u^2).$$

Move all terms to one side:

$$12.56u^2 - 40u + 17.56 = 0.$$

Solving gives approximately

$$u_1 \approx 0.52, \qquad u_2 \approx 2.66.$$

Therefore

$$\alpha_1 \approx \arctan(0.52) \approx 27.5^\circ, \qquad \alpha_2 \approx \arctan(2.66) \approx 69.4^\circ.$$

The target can be reached by a lower, flatter shot or by a higher, steeper shot.

15 A systematic problem-solving method

Recommended method

For almost every projectile motion problem, follow these steps:

1. Draw axes and mark the launch point, landing point, and target point.
2. Choose $y = 0$ clearly: ground level, launch level, or another convenient reference.
3. Resolve the initial velocity into components:

$$v_{0x} = v_0 \cos \alpha, \qquad v_{0y} = v_0 \sin \alpha.$$

4. Write the two coordinate equations:

$$x(t) = x_0 + v_{0x}t, \qquad y(t) = y_0 + v_{0y}t - \frac{1}{2}gt^2.$$

5. Use the condition given in the problem, such as $y = 0$, $x = X$, $v_y = 0$, or $y = Y$.
6. Solve for the required unknown.
7. Check units and whether the numerical answer is physically reasonable.

16 Common mistakes and how to avoid them

Mistake 1: using g with the wrong sign

If y is positive upward, then acceleration is

$$a_y = -g.$$

The vertical equation is

$$y(t) = y_0 + v_{0y}t - \frac{1}{2}gt^2.$$

Do not write a plus sign before the quadratic term unless you have chosen the downward direction as positive.

Mistake 2: assuming the flight time formula always works

The formula

$$t = \frac{2v_0 \sin\alpha}{g}$$

only holds when launch and landing heights are equal. If not, solve the vertical quadratic equation.

Mistake 3: saying that velocity is zero at the top

At the top,

$$v_y = 0,$$

but usually

$$v_x = v_0 \cos\alpha \neq 0.$$

So the velocity is not zero; it is horizontal.

Mistake 4: mixing degrees and radians

Trigonometric functions depend on angle units. A calculator set to radians will not interpret 30 as 30°. Always check the angle mode.

17 Practice problems

1. A projectile is launched from ground level at $v_0 = 16\,\mathrm{m/s}$ and $\alpha = 45°$. Find the time of flight, maximum height, and range.
2. A ball is thrown horizontally from a cliff of height $20\,\mathrm{m}$ with speed $10\,\mathrm{m/s}$. Find the time to hit the ground and the horizontal distance traveled.
3. A projectile is launched from ground level at $v_0 = 30\,\mathrm{m/s}$ and $\alpha = 60°$. Find its velocity vector at $t = 2\,\mathrm{s}$.
4. A projectile is launched from height $5\,\mathrm{m}$ with speed $14\,\mathrm{m/s}$ at 35°. Find the time of flight and range.

5. A projectile launched from the origin with speed 22 m/s must pass through the point (30 m, 4 m). Set up the quadratic equation for $\tan\alpha$.
6. For equal launch and landing height, prove that angles α and $90^\circ - \alpha$ give the same range.
7. Show that the maximum height can be obtained from the condition $v_y = 0$ or from energy conservation.
8. A projectile lands at the same height with impact speed 18 m/s. What was its launch speed in the ideal model? Explain.

Answers and hints

1. $t \approx 2.31\,\text{s}$, $H \approx 6.52\,\text{m}$, $R \approx 26.1\,\text{m}$.
2. $t = \sqrt{2h/g} \approx 2.02\,\text{s}$, $R \approx 20.2\,\text{m}$.
3. $v_x = 15.0\,\text{m/s}$, $v_y = 30\sin 60^\circ - 19.62 \approx 6.36\,\text{m/s}$.
4. Solve $0 = 5 + 14\sin 35^\circ t - 4.905t^2$; then use $R = 14\cos 35^\circ t$.
5. Use $4 = 30u - \frac{9.81\cdot 30^2}{2\cdot 22^2}(1+u^2)$, where $u = \tan\alpha$.
6. Use $\sin(2(90^\circ - \alpha)) = \sin(180^\circ - 2\alpha) = \sin(2\alpha)$.
7. Both methods give $H = v_0^2\sin^2\alpha/(2g)$ relative to launch height.
8. It was also 18 m/s, because mechanical energy is conserved and launch and landing heights are equal.

18 Formula sheet

Basic equations

$$x(t) = x_0 + v_0\cos\alpha\, t$$
$$y(t) = y_0 + v_0\sin\alpha\, t - \frac{1}{2}gt^2$$
$$v_x = v_0\cos\alpha$$
$$v_y = v_0\sin\alpha - gt$$

Trajectory

For $x_0 = 0$:

$$y(x) = y_0 + x\tan\alpha - \frac{g}{2v_0^2\cos^2\alpha}x^2$$

Maximum height

$$t_{\text{max}} = \frac{v_0\sin\alpha}{g}$$
$$H_{\text{max}} = y_0 + \frac{v_0^2\sin^2\alpha}{2g}$$

Same launch and landing height

$$t_{\text{flight}} = \frac{2v_0\sin\alpha}{g}$$
$$R = \frac{v_0^2\sin(2\alpha)}{g}$$
$$R_{\text{max}} = \frac{v_0^2}{g} \quad \text{at} \quad \alpha = 45^\circ$$

Launch from height y_0

If landing is at $y = 0$:

$$t_{\text{flight}} = \frac{v_0 \sin\alpha + \sqrt{v_0^2 \sin^2\alpha + 2gy_0}}{g}$$

$$R = v_0 \cos\alpha \, t_{\text{flight}}$$

Energy

$$\frac{1}{2}mv^2 + mgy = \frac{1}{2}mv_0^2 + mgy_0$$

$$v(y) = \sqrt{v_0^2 + 2g(y_0 - y)}$$

19 Summary

Projectile motion is a direct application of Newton's second law with a constant downward acceleration. The horizontal motion is uniform, while the vertical motion is uniformly accelerated. The two components are mathematically separate but physically simultaneous.

The most important equations are

$$x(t) = x_0 + v_0 \cos\alpha \, t, \qquad y(t) = y_0 + v_0 \sin\alpha \, t - \frac{1}{2}gt^2.$$

From them follow the parabolic trajectory, the time to reach the highest point, the maximum height, the flight time, the range, and the impact velocity. Most errors in projectile-motion problems come from using the equal-height formulas in situations where the heights differ, or from forgetting that only the vertical component of velocity is zero at the top.